

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 04

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Recap of Previous Lecture



Topic

Instruction Pipelining

Topic

Pipeline Hazards

Topic

Operand Forwarding

Topics to be Covered



Topic

Classical RISC Pipeline

Topic

Non-Linear Pipeline

#Q. An instruction format has the following structure

Instruction number: opcode destination reg, source reg-1, source reg-2

Consider the following sequence of instruction to be executed in a pipelined processor

I1: DIV R3, R1, R2	$R3 \leftarrow R1 / R2$
I2: SUB R5, R3, R4	$R5 \leftarrow R3 - R4$
I3: ADD R3, R5, R6	$R3 \leftarrow R5 + R6$
I4: MUL R7, R3, R8	$R7 \leftarrow R3 * R8$

Which of the following statement is/are TRUE?

- ☐ A There is a WAW dependency on R3 between I3 and I4
- ☒ B There is a WAW dependency on R3 between I1 and I3
- ☒ C There is a RAW dependency on R3 between I1 and I2
- ☐ D There is a RAW dependency on R3 between I2 and I3

#Q. Consider the following sequence of instructions, which is to be executed on a 5 stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding $5 + 6 - 1 = 10$
2. When pipeline does not use operand forwarding $10 + 2 * (5 - 3) = 14$

ADD R2, R1, R0

MUL R4, R3, R2

SUB R6, R5, R4

ADD R6, R7, R8

MUL R7, R1, R2

SUB R1, R3, R4

R2 ← R1 + R0

R4 ← R3 * R2

R6 ← R5 - R4

R6 ← R7 + R8

R7 ← R1 * R2

R1 ← R3 - R4

#Q. Consider the following sequence of instructions, which is to be executed on a 5 stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding $(5 + 6 - 1) + 2 = 12$
2. When pipeline does not use operand forwarding $(5 + 6 - 1) + 4 = 14$

ADD R2, R1, R0

R2 ← R1 + R0

MUL R4, R3, R2

R4 ← R3 * R2

LOAD R7, (1000)

R7 ← M[1000]

ADD R6, R7, R8

R6 ← R7 + R8

MUL R7, R1, R2

R7 ← R1 * R2

SUB R1, R3, R4

R1 ← R3 - R4



Topic : CPI Calculation in Pipeline

without hazard:-

for n instrs, cycles = $k+n-1$

$$\text{for 1 instr}^n, \text{---||---} = \boxed{\frac{k+n-1}{n}} \Leftarrow \text{CPI}$$

in ideal condition:-

$$\boxed{\text{CPI} = 1}$$

with hazard :-

no. of total stalls = x

$$CPI = \frac{k+n-1+x}{n}$$

in ideal condition $\Rightarrow$

$$CPI = \frac{n+x}{n} = 1 + \frac{x}{n}$$

$$\text{Avg. instr}^n \text{ execution time} = \text{CPI} * t_p$$

$$\text{without hazard} \quad S_{\text{ideal}} = \frac{t_n}{t_p}$$

$$\text{with hazard} \quad S_{\text{ideal}} = \frac{t_n}{\text{CPI}_{\text{avg}} * t_p}$$

#Q. Consider a 5-stage pipeline which is executing a program of 1000 instructions.

Among all instructions 200 instructions cause 2 stall cycles each.

1. Calculate CPI of pipeline? $\frac{5 + 1000 - 1 + 400}{1000} = 1.404$

2. If pipeline cycle time is 3ns then what is average instruction execution time? $1.404 * 3ns = 4.212 ns$

3. Calculate CPI of pipeline in ideal conditions? $1 + \frac{400}{1000} = 1.4$

4. If pipeline cycle time is 3ns then what is average instruction execution time in ideal conditions? $= 1.4 * 3ns = 4.2 ns$

total stalls = $200 * 2 = 400$

$$CPI_{ideal} = 1 + \frac{400}{1000}$$

$$= 1 + \frac{200 * 2}{1000}$$

$$= 1 + (0.2 * 2)$$

Ques) 20% inst^{ns} $\Rightarrow$ 2 stalls each

$$CPI_{ideal} = 1 + 0.2 * 2$$
$$= 1.4$$

#Q. Consider a 5-stage pipeline in which 30% instructions cause 1 stall cycle each. CPI of the pipeline is 1.3?

$$\begin{aligned}CPI &= 1 + 0.3 * 1 \\ &= 1.3\end{aligned}$$

#Q. Consider a 5-stage pipeline in which 20% instructions cause 2 stall cycles each and 16% instructions cause 1 stall cycle each. CPI of the pipeline is 1.56?

$$\begin{aligned} \text{CPI} &= 1 + (0.2 * 2) + (0.16 * 1) \\ &= 1.56 \end{aligned}$$

[NAT]

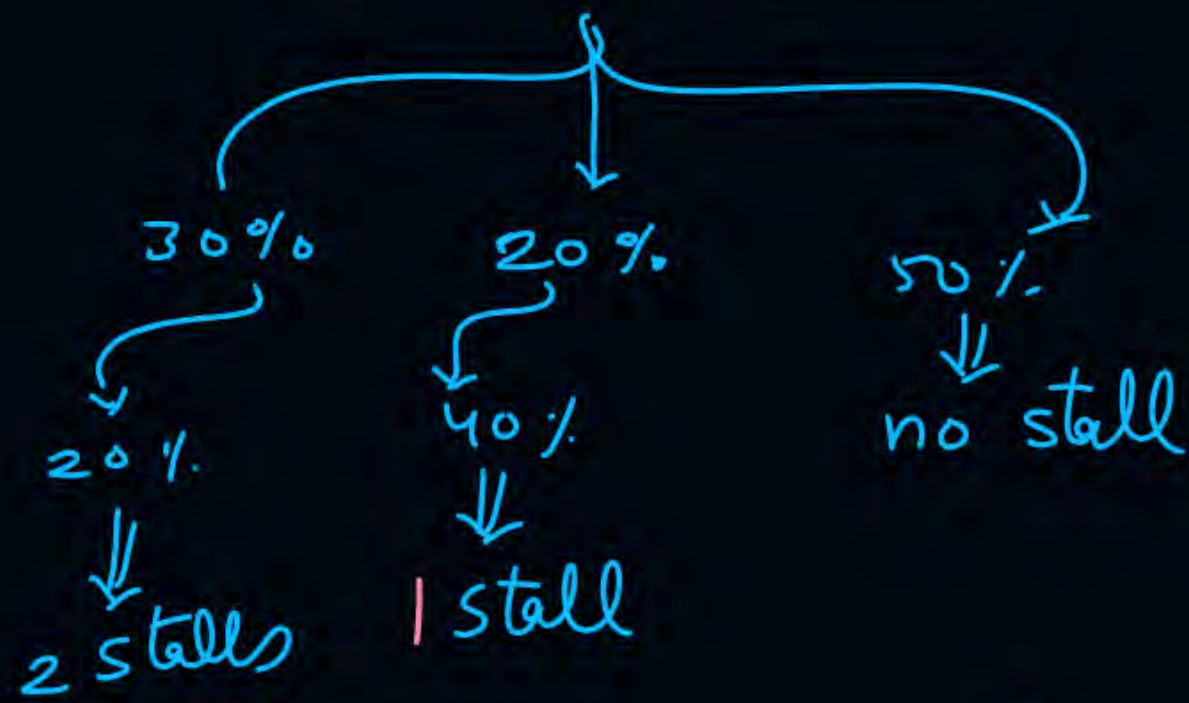
each stage delay = $x \Rightarrow t_p = x$



#Q. Consider a 5-stage instruction pipeline, where all stages take equal delay. When an application is executing on this 5-stage pipeline, consider 20% of the instructions incur 3 pipeline stall cycles is.

1. Average CPI for pipeline? $1 + 0.2 * 3 = 1.6$
2. Average instruction execution time for pipeline? $1.6 * x$
3. The speedup of pipeline achieved with respect to non-pipeline?

$$= \frac{t_n}{CPI * t_p} = \frac{5x}{1.6x} = \underline{\underline{3.125 \text{ Ans.}}}$$

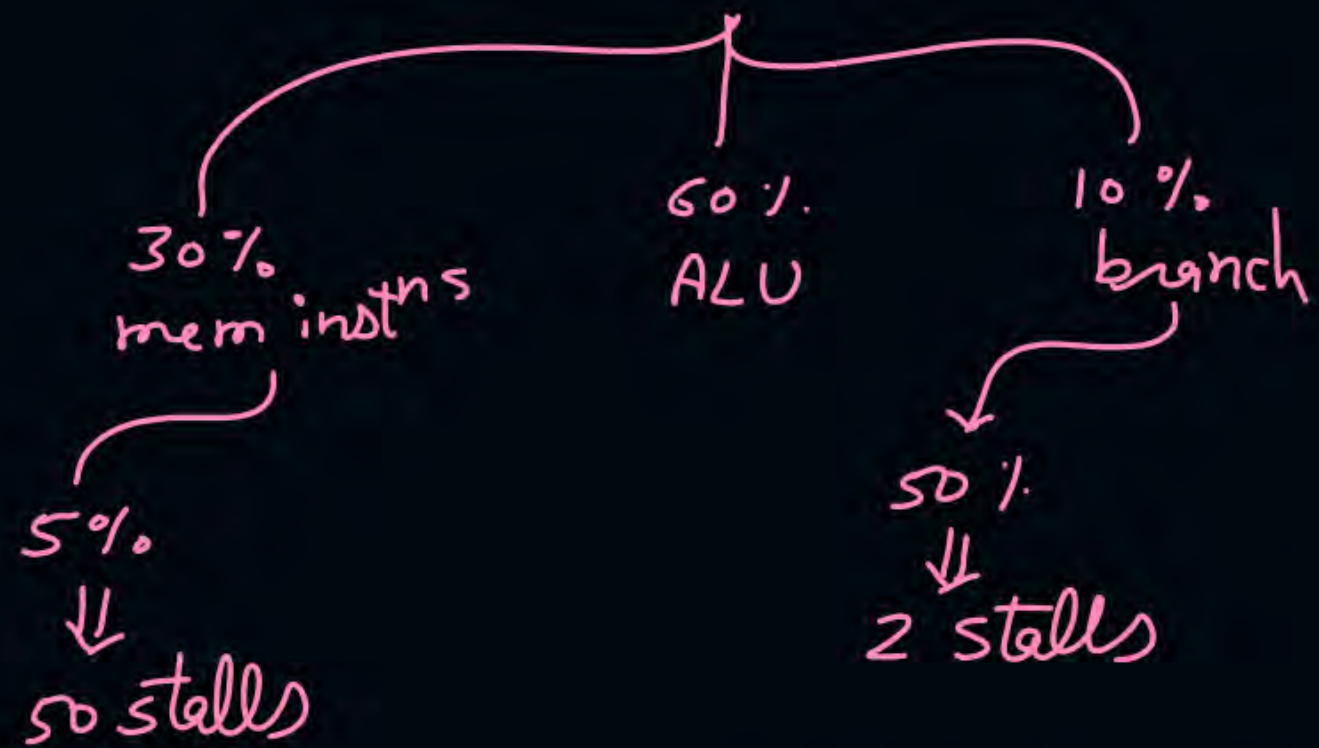


$$\begin{aligned}CPI &= 1 + (0.3 * 0.2 * 2) + (0.2 * 0.4 * 1) \\ &= 1.2\end{aligned}$$

$$\text{Ans} = 2.16$$

#Q. Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5-stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is 2.16.

$$t_n = \frac{1}{2.5 \text{ GHz}} * 5 = 2 \text{ ns} \quad \left| \quad t_p = \frac{1}{2 \text{ GHz}} = 0.5 \text{ ns}$$



$$= 1 + (0.3 * 0.05 * 50) + (0.1 * 0.5 * 2)$$
$$= 1.85$$

$$\text{speed up} = \frac{t_n}{\text{CPI} * t_p}$$
$$= \frac{2 \text{ ns}}{1.85 * 0.5 \text{ ns}}$$
$$= 2.162$$

Ans = 1.43

- #Q. A processor X_1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2 . If the BPU has a prediction accuracy of 80%, the speed up (rounded off to two decimal places) obtained by X_2 over X_1 in executing P is

X₁

$$CPI = 1 + 0.3 * 2 = 1.6$$

X₂

$$CPI = 1 + 0.3 * 0.2 * 2 = 1.12$$

$$= \frac{1.6 * 0.5m}{1.12 * 0.5m}$$

$$= 1.428$$

Efficiency of pipeline:-

$$\text{Efficiency} = \frac{\text{achieved speed up}}{\text{max speed up}} * 100\%$$

ex:- A pipeline can achieve a speed up 4 but max possible speed up is 5.

$$\begin{aligned}\text{efficiency} &= \frac{4}{5} * 100\% \\ &= 80\%\end{aligned}$$

$$\text{efficiency} = \frac{\frac{t_n}{CPI_{avg} * t_p}}{\frac{t_n}{t_p}} * 100\%$$

$$\text{efficiency} = \frac{1}{CPI_{avg}} * 100\%$$

ques) assume a k stage pipeline where all stages take equal delay.
if pipeline can achieve a speed up of 3. and efficiency of 75% then value of k 4?

Solⁿ

if all stage delay = x

$$t_n = kx$$

$$t_p = x \quad S_{max} = k$$

$$3 = 0.75 * k$$

$$k = \frac{3}{0.75}$$

$$k = 4$$



Topic : Classic RISC Pipeline

1. IF — Instⁿ fetch
2. ID — Instⁿ decode and reg. read
3. EX — Execution of ALU operation
4. MEM — mem. access
5. WB — Reg. write



Topic : Classic RISC Pipeline for Computation

1. IF — $inst^n$ fetch & PC $inst^n$
2. ID — $inst^n$ decode | operand Read from reg.
3. EX — ALU operation
4. MEM — ∞
5. WB — write back to Reg.

no. of stalls for data hazard

$$= 5 - 2$$

$$= 3$$



Topic : Classic RISC Pipeline for Load

1. IF — $Inst^n$ fetch & PC inc.

2. ID — $Inst^n$ decode

3. EX — X

4. MEM — mem. Read

5. WB — Reg. write



Topic : Classic RISC Pipeline for Store

1. IF — $Inst^n$ fetch & PC inc.
2. ID — $Inst^n$ decode | Reg. Read (operand fetch from Reg.)
3. EX — X
4. MEM — mem. write
5. WB — $\varnothing$



Topic : Classic RISC Pipeline for Branch

1. IF — Instⁿ fetch

2. ID — Instⁿ decode | Effective add. calculation | Condition check | PC update

3. EX — X

4. MEM — X

5. WB — X

no. of stalls for branch hazard = $2 - 1$
= 1



Topic : Non-Linear Pipeline



instrs may need stages multiple times at different times
during execution.



Topic : Non-Linear Pipeline

reservation table

← no. of cycles →

stages

	1	2	3	4	5	6	7	8
S1	X					X		X
S2		X		X				
S3			X		X		X	

		1	2	3	4	5	6	7	8	9	10	11	12
I1		S1	S2	S3	S2	S3	S1	S3	S1				
I2			S1	S2	S3	S2	S3	S1	S3	S1			
I3				S1	-	-	S2	-	-	S3			



Topic : Non-Linear Pipeline

Latency: The number of time units (clock cycles) between two initiations of a pipeline is the latency between them. Latency values must be non-negative integers.

→ Resource conflict / structural hazard
Collision: When two or more initiations are done at same pipeline stage at the same time will cause a collision.

Problem statement :- finding latency w/o any collision.



Topic : Non-Linear Pipeline

Forbidden Latency: Latencies that cause collisions are called forbidden latencies.

$$\begin{array}{l|l|l} \underline{S1}:- & (6-1) = 5 & \underline{S2}:- (4-2) = 2 \\ & (8-1) = 7 & \underline{S3}:- (5-3) = 2 \\ & (8-6) = 2 & (7-3) = 4 \\ & & (7-5) = 2 \end{array} \quad \{2, 4, 5, 7\}$$

Permissible Latency: Latencies that do not cause any collision are called permissible latencies $\{1, 3, 6, 8\}$



Topic : Non-Linear Pipeline

Collision Vector:

$$8 \text{ cycles} \Rightarrow C_8 C_7 C_6 C_5 C_4 C_3 C_2 C_1 = 01011010$$

$$C_i \begin{cases} 0 & \text{for permissible latency} \\ 1 & \text{for forbidden latency} \end{cases}$$



Topic : Non-Linear Pipeline

For every zero in collision vector at position i :

1. Shift the state vector right i positions
2. Take bit-wise OR of this with original collision vector
3. Draw an edge from the state to new state of OR result with i as latency label

state $\Rightarrow 01011010$:-

0 at position 1 $\Rightarrow$

$$\begin{array}{r} 01011010 \\ \swarrow \quad \searrow \times \\ \text{OR} \begin{array}{r} 00101101 \\ 01011010 \\ \hline 01111111 \end{array} \end{array}$$

0 at position 3 $\Rightarrow$

$$\begin{array}{r} 01011010 \\ \swarrow \quad \searrow \times \\ \text{OR} \begin{array}{r} 00001011 \\ 01011010 \\ \hline 01011011 \end{array} \end{array}$$

0 at position 6 $\Rightarrow$

$$\begin{array}{r} 01011010 \\ \swarrow \quad \searrow \times \\ \text{OR} \begin{array}{r} 00000001 \\ 01011010 \\ \hline 01011011 \end{array} \end{array}$$

0 at position 8 $\Rightarrow$

$$\begin{array}{r} 00000000 \\ \text{OR} \begin{array}{r} 01011010 \\ \hline 01011010 \end{array} \end{array}$$

state 01111111 :-

0 at position 8 $\Rightarrow$

$$\begin{array}{r} 00000000 \\ \text{OR} \begin{array}{r} 01011010 \\ \hline 01011010 \end{array} \end{array}$$

state 01011011 =>

0 at position 3 :-

00001011
OR 01011010

01011011

0 at position 6 :-

00000001
OR 01011010

01011011

0 at position 8 :-

00000000
OR 01011010

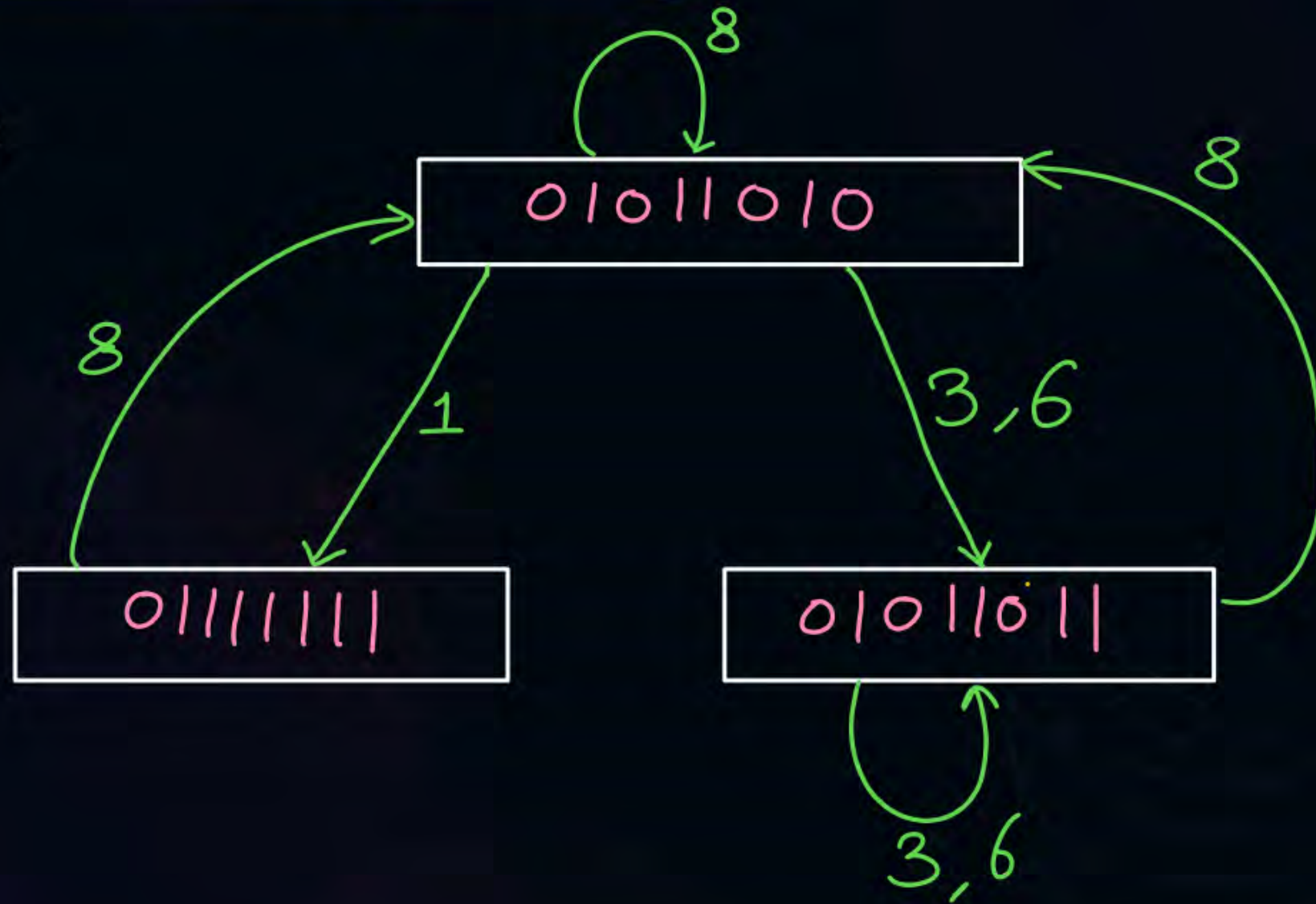
01011010



Topic : Non-Linear Pipeline



State Diagram:





Topic : Non-Linear Pipeline

Simple Cycle:

A Simple Cycle is a latency cycle in which each state appears only once, except the starting state.

$$\{8\}, \{1, 8\}, \{3, 8\}, \{6, 8\}, \{3\}, \{6\}$$

Greedy Cycle:

A Greedy Cycle is a simple cycle whose edges are all made with minimum latencies from their respective starting states.

$$\begin{array}{l|l} \{1, 8\} & \{3\} \\ \hline \text{avg} = \frac{1+8}{2} & \text{avg} = 3 \\ = 4.5 & \end{array}$$

MAL (Minimum Average Latency): $\min(4.5, 3) = 3$

#Q. Consider the following reservation table for a pipeline having three stages S1, S2 and S3.

Time →					
	1	2	3	4	5
S1	X				X
S2		X		X	
S3			X		

The minimum average latency (MAL) is _____?

#Q. Consider a program which contains 50 instructions I1, I2 , I3 I50. Further consider a 5-stage classical RISC pipeline with stages as: IF, ID, EX, MEM and WB. The program contains only 1 branch instruction which is instruction I5, and its target is instruction I48. If during the execution of the program the branch is taken, then number of cycles required to execute this program in the given pipeline is 13 ?

$$n = 5 + 3 = 8$$

$$\begin{aligned} \text{w/o hazard} &= 5 + 8 - 1 = 12 \\ \text{stalls} &= (2 - 1) = 1 \\ \hline \text{Total} &= 13 \end{aligned}$$

[NAT]

$$\text{Ans} = \underline{\underline{1.88}}$$



#Q. Consider a pipelined processor with 5 stages, Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB). Each stage of the pipeline, except the EX-stage, takes one cycle. Assume that the ID stage merely decodes the instruction and the register read is performed in the EX-stage. The EX-stage takes one cycle for ADD instruction and two cycles for MUL instruction. Ignore pipeline register latencies.

Consider the following sequence of 8 instructions:

ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL

Assume that every MUL instruction is data-dependent on the ADD instruction just before it and every ADD instruction (except the first ADD) is data-dependent on the MUL instruction just before it. The Speedup is defined as follows:

$$\text{Speed up} = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forwarding}} = \frac{30}{16} = 1.875 = 1.88$$

The Speedup achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is ____.

IF — 1		
ID — 1	ADD	MUL
EX — 1	1	2
MEM — 1	stall	1
WB — 1		

with operand fwd.:-

$$\text{w/o stall} = 5 + 8 - 1 = 12$$

$$\text{structural hazard stalls} = 4 * 1 = 4$$

$$\text{Total} = 16$$

w/o op. fwd.

$$\text{w/o hazard} = 5 + 8 - 1 = 12$$

$$\text{structural hazard stalls} = 4$$

$$\text{Data} \quad \text{---} 11 \text{---} = (5 - 3) * 7 = 14$$

$$\text{Total} = 30$$



2 mins Summary



Topic

Classical RISC Pipeline

Topic

Non-Linear Pipeline





Happy Learning

THANK - YOU

